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Research paper

Design, synthesis and biological evaluation of novel 2,3-dihydroquinazolin- 4(1*H*)-one derivatives as potential fXa inhibitors

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ABSTRACT

Coagulation factor Xa (fXa) is a particularly attractive target for the development of effective and safe anticoagulants. In this study, novel 2,3-dihydroquinazolin-4(1*H*)-one derivatives were designed as potential fXa inhibitors based on anthranilamide structure which has been reported in our previous research. The experimental data showed that most of the designed compounds exhibited significant *in vitro* potency against fXa. Among them, compound **8e** displayed the strongest potency against fXa with the IC_{50} value of 21 nM and highly selectivity versus thrombin (IC_{50} = 67 μM) and excellent *in vitro* antithrombotic activity with its 2 × PT value of 1.2 μM and 2 × aPTT value of 0.6 μM. In addition, **8e** also displayed excellent *in vivo* antithrombotic activity in the rat arteriovenous shunt (AV-SHUNT) model. The bleeding risk evaluation showed that **8e** had a similar safety profile as that of betrixaban. All results demonstrated that compound **8e** could be considered as a potential fXa inhibitor for the prevention and treatment of thromboembolic diseases.

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1. Introduction

Thrombotic disorders, including ischemic stroke, unstable angina, pulmonary embolism (PE), deep venous thrombosis (DVT) and myocardial infarction, constitute a major cause of morbidity and mortality worldwide [1,2]. Although several anticoagulants such as fondaparinux, warfarin, low molecular weight heparins (LMWHs) and unfractionated heparin are effective in the prevention and treatment of these thrombotic diseases, they also display several undesirable effects [3–5]. In the past decades, numerous research efforts have been made to identify novel anticoagulants with improved efficacy and safety. After extensive investigation, thrombin (fIIa) and fXa have become the promising targets for the development of potent, selective, and orally bioavailable anticoagulants [6–8].

Activated serine protease fXa, which is situated at the common pathway of coagulation cascade, catalyzes the conversion of prothrombin to thrombin and plays a central role in various thromboembolic disorders [9,10]. Compared with thrombin inhibitors, selective fXa inhibitors have better specific antithrombotic effects and lower risk of bleeding due to their ability to reduce the further generation of thrombin without affecting the existing level of thrombin [11]. Currently, there are three oral, direct and selective fXa inhibitors approved by the Food and Drug Administration (FDA): rivaroxaban [12,13], apixaban [14,15] and edoxaban [16] (Fig. 1). These novel fXa inhibitors exhibit advantages compared with traditional anticoagulants. However, they still have several disadvantages with the most notable being their narrow indications for use and the lack of an effective antidote for when significant bleeding occurs [17–21]. The development of an antidote for use with fXa inhibitors is being investigated and would address some of the concerns with the fXa inhibitors, but it would not address the safety and efficacy of the inhibitors themselves. [22–24], Therefore, the further development of novel and safer fXa inhibitors is desirable to increase use of fXa inhibitors in the clinic.

Most fXa inhibitors bind to the active site in a characteristic L-shaped conformation [8]. They have a three-component system which includes a central scaffold and two hydrophobic fragments

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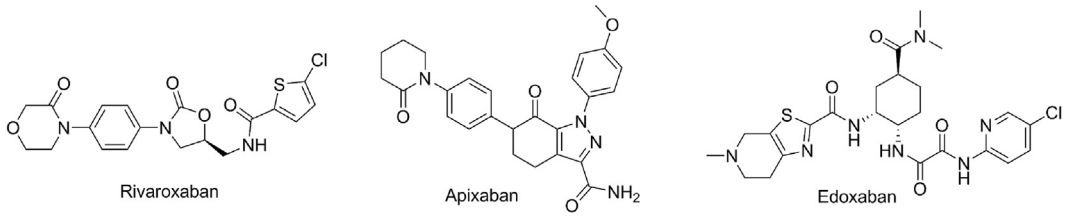


Fig. 1. The structures of Rivaroxaban, Apixaban and Edoxaban.

(*P*1 and *P*4) that provide a non-linear geometry considered to play a crucial role in fXa recognition [25]. The core linkers studied to date can be classified into either highly flexible class or fairly rigid class. In our previous study, we have identified a series of anthranilamide derivatives as fXa inhibitors by a fragment-based strategy integrated with structure-based key pharmacophore and structural consensus docking [26]. Based on the anthranilamide scaffold and structure–activity analysis, novel 2,3-dihydroquinazolin-4(1*H*)-one as a fairly rigid linker was designed and its derivatives were synthesized and evaluated for *in vitro* and *in vivo* antithrombotic potency. Additionally, the compounds reported in this study have a similar structure to anti-tumor molecules designed by our group (unpublished results). Cancer is known to induce a hypercoagulable state, which is a major risk factor for VTE [27–30]. Thus, compounds possessing both anticoagulant and antitumor activities are desirable. Although this report focused on assessing the anticoagulant properties of these molecules, their anti-tumor activity is also being evaluated by our lab and will be reported in due course.

2. Drug design

Compound **X007** with pronounced fXa inhibitory activity has been reported in our published research [26]. The interaction mode of **X007** with fXa predicted by glide in Schrodinger 2009 which has been demonstrated to successfully predict the binding modes of many reported inhibitors to fXa is illustrated in Fig. 3A (PDB code: 2w26). 4-chlorobenzamido occupies the hydrophobic S1 pocket formed by Ala190, Val213 and Tyr228 and forms a Cl–π bond with the aromatic ring of Tyr228 which plays a significant role in the interaction between inhibitors and fXa [31–33]. The NH of the scaffold carboxamide forms a hydrogen bond with the carbonyl oxygen of Gly219. The *P*4 phenyllactam is flanked by the phenyl groups of Phe174 and Tyr99 in the S4 aryl binding pocket and forms an edge to face interaction with Trp215. The carbonyl oxygen in the central scaffold also forms a hydrogen bond with Gly216.

In order to explore novel scaffold of fXa inhibitors, compound **X138** was derived through the cyclization of **X007**. The docking study for **X138** (Fig. 3B) showed no hydrogen bond interaction between ligand and receptor. The *P*4 phenyllactam fits well into the hydrophobic S4 with a similar conformation to that of compound **X007**. The 4-chlorobenzamido is orientated to the S_1 region but is not able to occupy the S1 pocket very well due to the lack of Cl–π interaction. The dock results suggest that the conformations of 4-chlorobenzamido between the compound **X007** and **X138** have been changed as a result of the cyclization. For optimal the binding conformation of compound **X138** to fXa, the orientation of *P*4 needs to be modified and the length also needs to be extended to form the key Cl–π interaction. Thus, compound **X254** was designed and its binding mode with fXa was also calculated (Fig. 2 and 3c). The docking results exhibited that both *P*1 and *P*4 binding conformations were similar to that of **X007** and the Cl–π bond was also formed, which is considered to be important for the fXa inhibitory activity.

To confirm the reliability of the strategy, the two lead compounds **X138** and **X254** were synthesized and evaluated for *in vitro* inhibition of fXa. The experimental data showed that the two compounds exhibited inhibitory activity against fXa with the IC_{50} values of 52 μM and 127 nM, respectively. Encouraged by these results, a series of 2,3-dihydroquinazolin -4(1*H*)-one derivatives were then designed, synthesized and evaluated for their fXa inhibitory activity.

3. Results and discussion

3.1. Chemistry

3.1.1. General chemistry

The synthesis of intermediates and target compounds **X138**, **X254**, **6a**–**e**, **7a**–**f** and **8a**–**h** were given in Scheme 1. The intermediate **2** was formed by the copper-catalysed coupling reaction of 2-Piperidinone or morpholin-3-one with commercially available **1** in the presence of 8-hydroxyquinoline as a ligand [34,35]. Then treatment of compound **2** with commercially available 5-methoxy-2-nitrobenzoic acid provided intermediate **3**, which was reduced by using Pd/C (10%) and hydrazine hydrate (80%) in anhydrous EtOH to be converted to **4**. The cyclization of intermediate **4** with 37% aqueous H_2CO provided the key intermediate **5** [36], which was acidylated by aromatic carboxylic acids to afforded **6a**–**e** and **X138**. Alternatively, treatment of compound 5 with aromatic acetic acids provided **X254, 7a**–**f** and **8a**–**h**. The synthetic method for the preparation of compound **13** was outlined in Scheme 2, which was the same as that depicted in Scheme 1.

3.2. Biological activities and discussion

3.2.1. In vitro fXa inhibition activities studies

All the targeted compounds were evaluated *in vitro* for their fXa enzyme inhibitory activity, and betrixaban was selected as the positive control. The experimental results of **6a**–**e** and **7a**–**f** were summarized in Table 1. As expected, the analogs **6a**–**e** showed decreasing *in vitro* FXa inhibitory activities compared with those of **7a**–**f**. This demonstrated that the cyclization of anthranilamide disturbed the favorable arrangement of *P*1 conformations. As for **7a**–**f**, minor changes of *P*1 surrogates caused a considerable loss of potency compared with **X254**. For example, 3-chlorophenyl derivative **7d** and 2-chlorophenyl derivative **7e** were 3- and 7-fold less potent than that of **X254**, respectively. In addition, no substituent in the phenyl ring (**7a**) and the introduction of methoxy or acetamido into the 4-position of the phenyl ring decreased inhibitory potency (**7b**, **f** vs **X254**). These results suggested that the Cl–π interaction formed in the S1 pocket is essential for activity.

Structural modifications were made to *P*4 fragments with 4-chlorophenyl as the *P*1 substituent. Since morpholin-3-one and 2-Pyridone groups can serve as *P*4 surrogates to provide an excellent effect against fXa [32,37,38], compounds **8c** and **13** were designed, synthesized and evaluated for fXa inhibitory activity. As

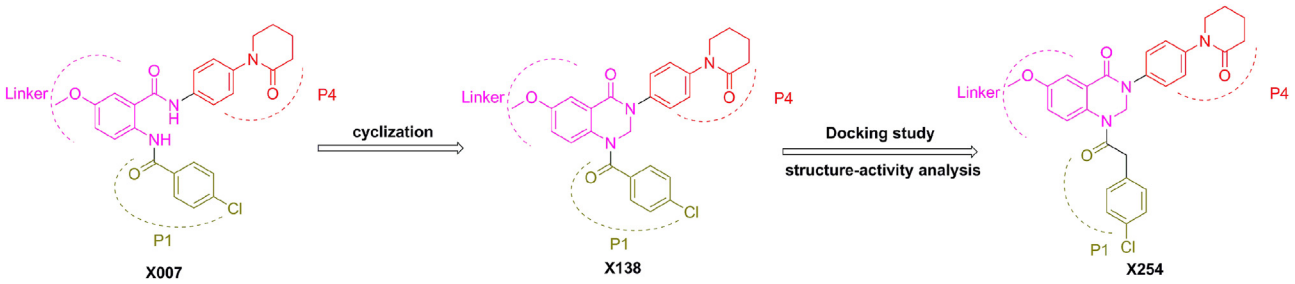


Fig. 2. Design of novel 2,3-dihydroquinazolin-4(1*H*)-one derivatives **X138** and **X254**.

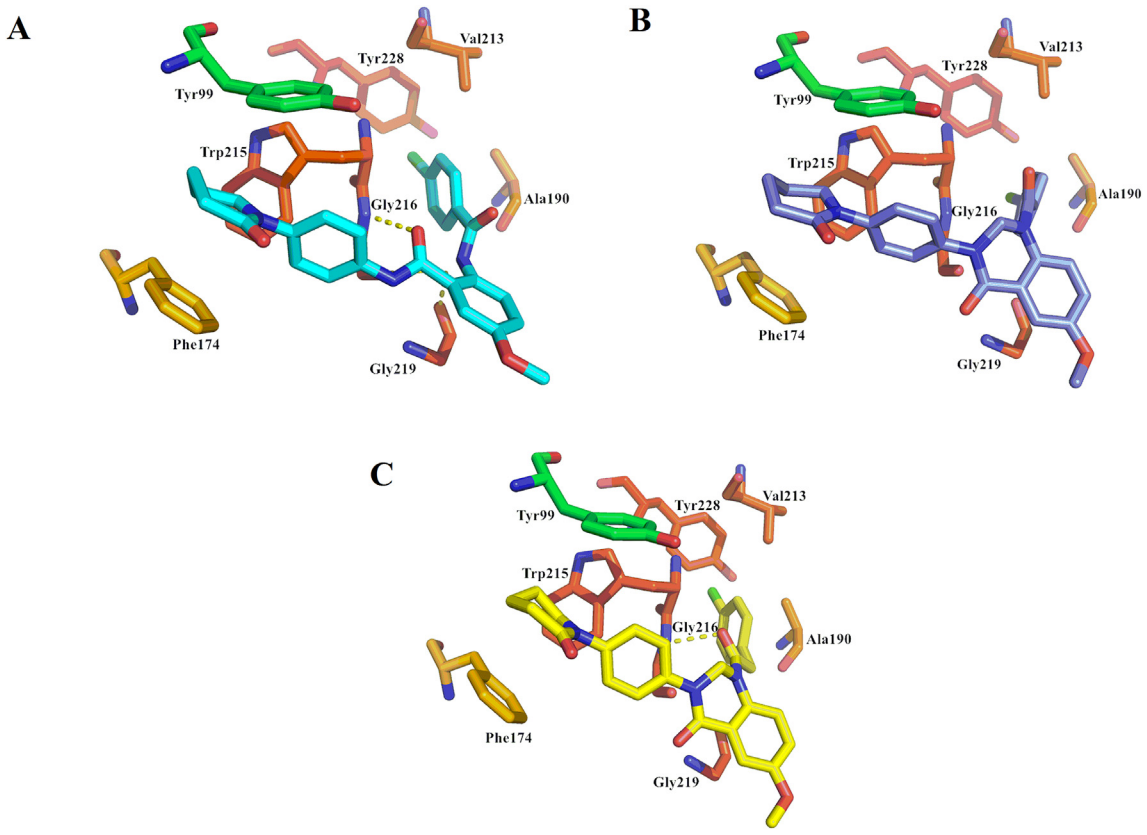


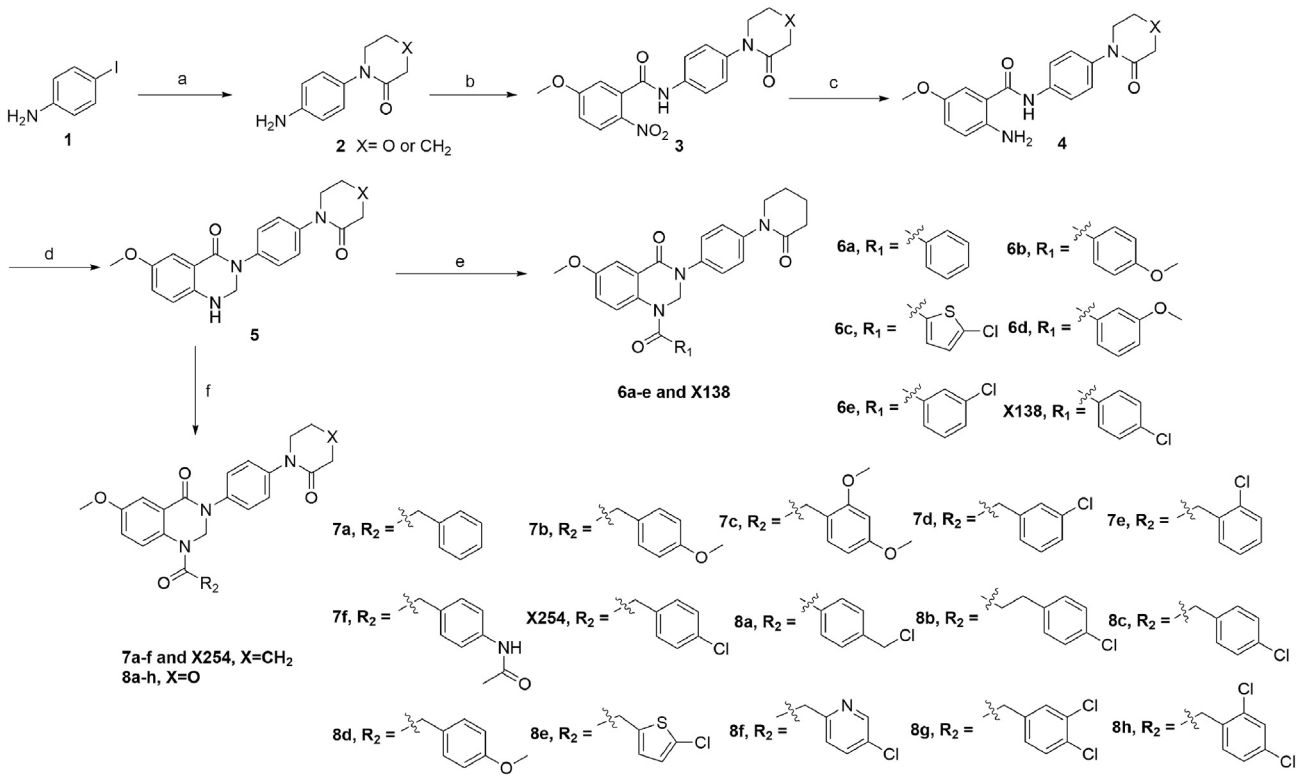
Fig. 3. The docking model of compounds **X007**, **X138** and **X254** with the binding sites of fXa (PDB code 2w26). (A): The interactions of compound **X007** with the active site of fXa. (B): The binding mode of compound **X138** with the active site of fXa. (C): The interactions of compound **X254** with the active site of fXa. Hydrogen atoms have been hidden for clarity. The figures were prepared using PyMol (www.pymol.org).

shown in Table 2, compound **8c** exhibited higher potency than **X254** while inhibitory activity of **13** was similar to that of **X254**.

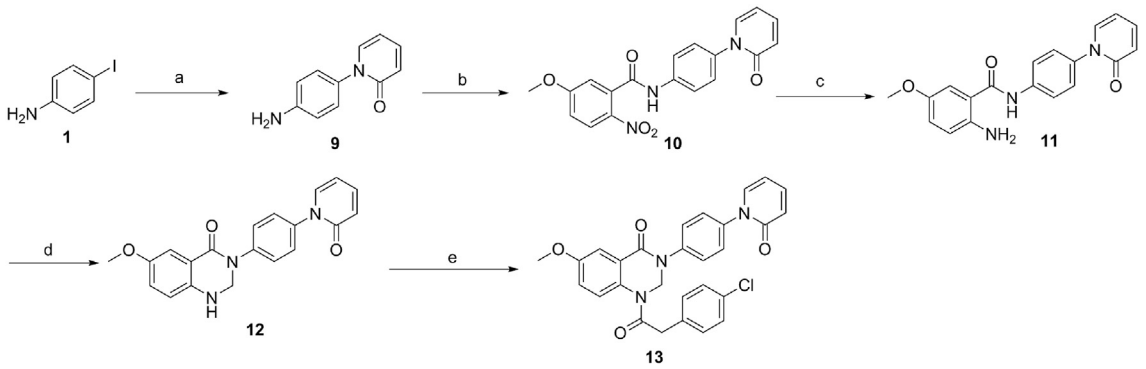
Based on these results, the 3-morpholinone as *P*4 substituent was kept constant while *P*1 analogs bearing different numbers and positions of Cl atoms were examined. Among them (**8a**–**h**), compound **8e** with 5-chlorothiophen as the *P*1 surrogate showed a 2-fold increase of activity compared with **8c** (Table 3). This was quite different from structure–activity relationship in **6c** and **6e**. Other modifications were to introduce one methylene between phenyl ring and Cl atom (**8a**) or methylene (**8b**), which caused a considerable decrease in potency. Replacement of the Cl atom in **8c** with a methoxyl group provided compound **8d** which was 17-fold less potent in the binding activity compared with that of **8c**. This result demonstrates that the Cl–π interaction formed by the chlorine atom and the phenyl ring of Tyr228 plays a crucial role in potency. The introduction of a second chlorine at the 2 or 3-position of phenyl ring caused a significant loss in potency (**8g**, **h** vs **8c**). The 5-chloropyridin derivative **8f** was also less active than **8c**.

3.2.2. *Selectivity versus thrombin and prothrombin time (PT) assay*

Given their excellent fXa inhibitory activity, compounds **8c** and **8e** were selected to assess their selectivity versus thrombin, the extension of the prothrombin time (PT) and activated partial thromboplastin time (aPTT) to further determine its *in vitro* anticoagulant activity. PT represents the effect of a molecule on the extrinsic pathway of coagulation, whereas aPTT measures the effect on the intrinsic pathway. As shown in Table 4, both compounds **8c** and **8e** displayed a high degree of selectivity against thrombin with IC_{50} values of 81 μM and 67 μM, respectively. Additionally, **8c** showed moderate *in vitro* anticoagulant activity judged by its 2 × PT (11.6 μM) and 2 × aPTT (16.1 μM) values. For its counterpart, **8e** exhibited excellent *in vitro* anticoagulant activity with 2 × PT



Scheme 1. Synthesis routes and structures of intermediates and target compounds. (a) 2-Piperidinone or morpholin-3-one, CuI, 8-hydroxyquinoline, K_2CO_3, DMSO, 130 °C, under N_2, 12 h; (b) Oxalyl chloride, DMF, CH_2Cl_2, 0-r.t., 1 h, then CH_2Cl_2, triethylamine, 0-rt, 16 h (c) hydrazine hydrate (80%), Pd/C (10%), EtOH, reflux, 2 h; (d) 37% aqueous H_2CO, NaOH, EtOH/H_2O, 37 °C, 15 min; (e) aromatic carboxylic acid, Oxalyl chloride, DMF, CH_2Cl_2, 0-rt, 1 h, then CH_2Cl_2, triethylamine, 0-rt, 14 h; (f) EDCI, DMAP, aromatic carboxylic acid, DMF, rt, 24 h.



Scheme 2. Synthesis routes and structures of intermediates and target compound **13**. (a) 2-Pyridone, CuI, 8-hydroxyquinoline, K_2CO_3, DMSO, 130 °C, under N_2, 12 h; (b) Oxalyl chloride, DMF, CH_2Cl_2, 0-r.t., 1 h, then CH_2Cl_2, triethylamine, 0-rt, 16 h (c) hydrazine hydrate (80%), Pd/C (10%), EtOH, reflux, 1 h; (d) 37% aqueous H_2CO, NaOH, EtOH/H_2O, 37 °C, 15 min; (e) aromatic carboxylic acid, Oxalyl chloride, DMF, CH_2Cl_2, 0-rt, 1 h, then CH_2Cl_2, triethylamine, 0-rt, 14 h.

value of 1.2 μM and 2 × aPTT value of 0.6 μM, which provided evidence that **8e** showed better *in vitro* anticoagulant activity and similar selectivity against thrombin compared with betrixaban.

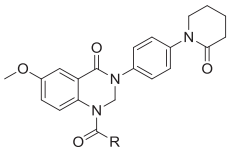
3.2.3. *In vivo antithrombotic activity in rats and tail-bleeding time effect of **8e** in rats*

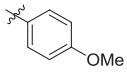
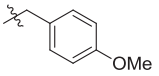
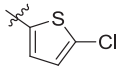
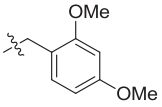
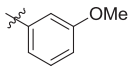
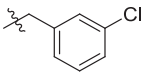
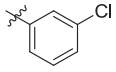
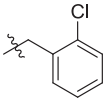
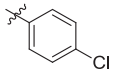
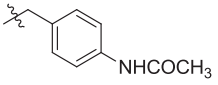
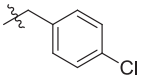
Based on comprehensive analyses of *in vitro* potency and selectivity, compound **8e** was selected to evaluate its *in vivo* antithrombotic efficacy in arteriovenous shunt (AV-SHUNT) thrombosis model in rats and bleeding risk in a rat tail-bleeding time study. As shown in Fig. 4A, **8e** displayed potent *in vivo* antithrombotic activity and reduced thrombus weight in a dose-dependent manner with a significant reduction observed post-dosing of 10 mg/kg, which is similar to that of betrixaban. Furthermore, **8e** and betrixaban prolonged bleeding time in a dose-dependent manner to the same extent, suggesting that they have a similar safety degree in terms of bleeding risk (Fig. 4B).

3.3. *Docking simulation for compound **8e***

To gain insight into the binding model of compound **8e** to fXa, docking studies were carried out using glide in Schrodinger 2009 with the crystal structure of fXa as PDB code 2w26. The results were shown in Fig. 5. The carbonyl oxygen of acetyl in the *P*1 fragment

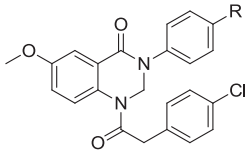
Table 1
fXa inhibitory effect of designed compounds **X254**, **X138**, **6a**–**e** and **7a**–**g**.

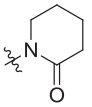
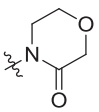
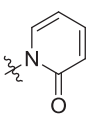


Compd.	R	IC_{50} (μM)[a]	Compd.	R	IC_{50} (nM)[a]
6a		546 ± 127	**7a**		4065 ± 1078
6b		113 ± 21	**7b**		887 ± 234
6c		179 ± 8	**7c**		1267 ± 349
6d		354 ± 65	**7d**		532 ± 112
6e		243 ± 39	**7e**		1345 ± 356
X138		52 ± 11	**7f**		234 ± 52
Betrixaban	_	4.5 (nM)	**X254**		127 ± 25

[a] The data represent the mean of at least three independent determinations.

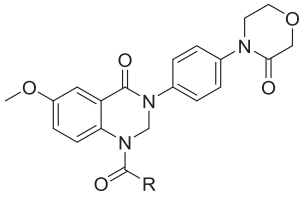
Table 2
fXa inhibitory effect of designed compounds **X254, 8c** and **13**.

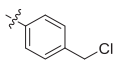
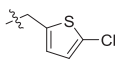
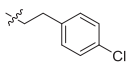
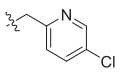
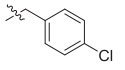
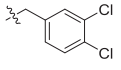
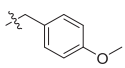
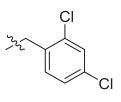


Compd.	R	IC_{50} (nM)[a]
X254		127 ± 25
8c		42 ± 9
13		165 ± 38

[a] The data represent the mean of at least three independent determinations.

Table 3
fXa inhibitory effect of designed compounds **8a**–**h**.



Compd.	R	IC_{50} (nM)[a]	Compd.	R	IC_{50} (nM)[a]
8a		1414 ± 267	**8e**		21 ± 6
8b		2212 ± 455	**8f**		157 ± 42
8c		42 ± 9	**8g**		345 ± 65
8d		717 ± 198	**8h**		498 ± 96

[a] The data represent the mean of at least three independent determinations.

Table 4
Selectivity versus thrombin and anticoagulant activity of **8c** and **8e**.

Compd.	Thrombin IC_{50}(μM)[a]	2 × PT (μM)[b]	2 × aPTT (μM)[c]
8c	81	11.6	16.1
8e	67	1.2	0.6
Betrixaban	18	2.4	1.1

[a] The data represent the mean of at least three independent determinations.
[b] PT values are defined as the inhibitor concentration required to double fibrin formation the time and using rabbit plasma.
[c] aPPT values are defined as the inhibitor concentration required to double activated partial thromboplastin time using rabbit plasma.

forms a strong hydrogen bond (2.6 Å) with the NH of Gly216. Morpholinone occupied the S4 pocket and kept a perpendicular arrangement to the nonpolar aryl ring which formed the pi-pi interaction with Trp215. 5-chlorothiophene of **8e** occupied hydrophobic S1 pocket and also forms the key Cl–π interaction with the aromatic ring of Tyr228 located at the bottom of the S1 pocket. In addition, the central 2,3-dihydroquinazolin-4(1*H*)-one provides the crucial L-shape needed for fXa binding. These results demonstrate that **8e** interacts with the fXa active site in a highly complementary manner.

4. Conclusion

In summary, two novel 2,3-dihydroquinazolin-4(1*H*)-one derivatives **X138** and **X254** were designed and synthesized based on the anthranilamide derivative **X007**. Then modifications focusing on the *P*1 and *P*4 groups resulted in a novel series of 2,3-dihydroquinazolin-4(1*H*)-one derivatives. *In vitro* fXa inhibitory activity studies demonstrated that most of the test compounds displayed some degree of fXa inhibition. Compound **8e** showed potent inhibitory activity with IC_{50} value 21 nM and excellent selectivity over thrombin (67 μM). Moreover, compound 8e also exhibited pronounced *in vitro* anticoagulant activity with its 2 × PT value of 1.2 μM and 2 × aPTT value of 0.6 μM, suggesting that **8e** could have a potent effect on both extrinsic and intrinsic pathways. *In vivo* bioactivity evaluation by the model of AV-SHUNT in rats showed **8e** had similar efficacy to betrixaban. Eventually, **8e** and betrixaban prolonged the bleeding time in a dose-dependent manner to the same extent in a rat tail-bleeding time study which exhibited that they have a similar safety profile in terms of bleeding risk. These results demonstrated that compound **8e** is a potent and selective fXa inhibitor and warrants further evaluation as a leading candidate for the prevention and treatment for

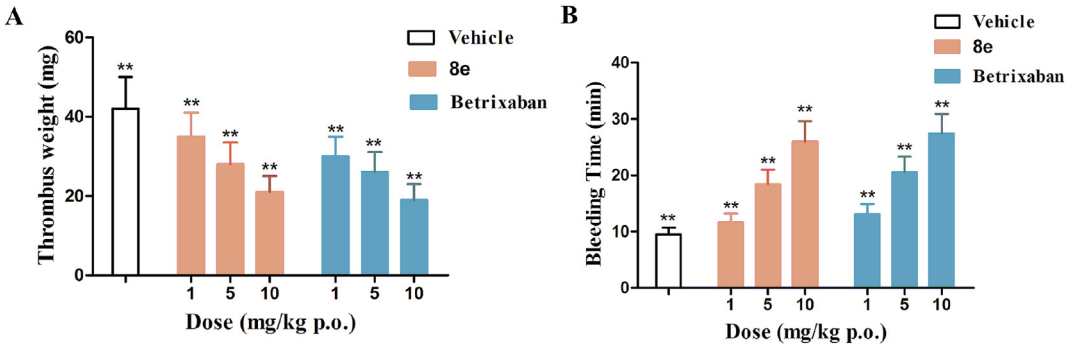


Fig. 4. (A): Antithrombotic profile of **8e** and betrixaban in the rat arteriovenous shunt model. (B) Effect of **8e** and betrixaban on rat tail-bleeding time. Values are mean ± SEM (n = 8). *P < 0.05, **P < 0.01, and ***P < 0.001 compared to vehicle group by Student's *t*-test.

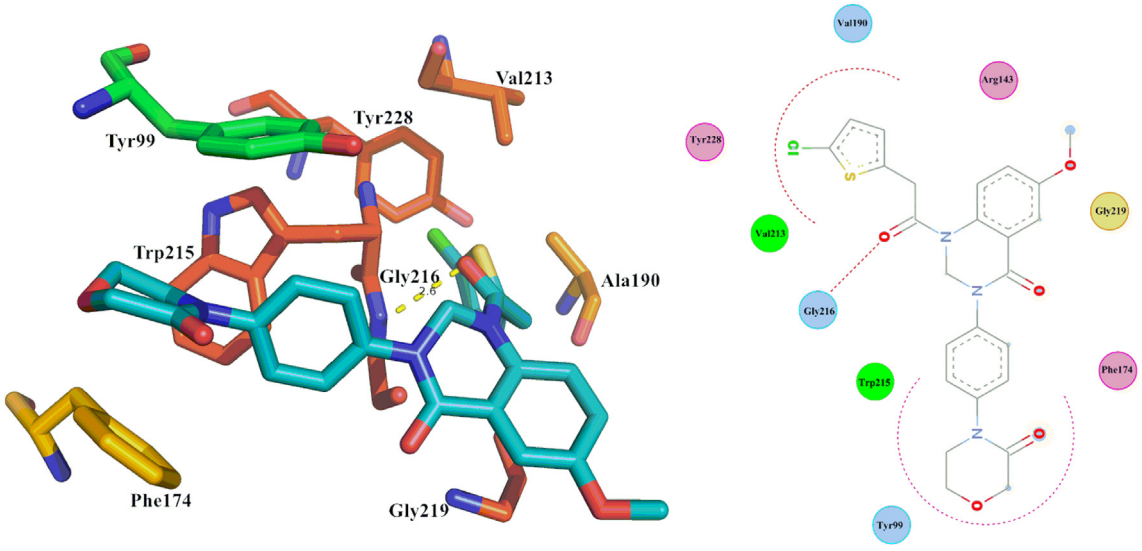


Fig. 5. The docking model of compound **8e** with the binding sites of fXa (PDB code 2w26). (A): 3D interactions of **8e** with the active sites of fXa. (B): 2D interactions of **8e** on the active site of FXa. Hydrogen atoms have been hidden for clarity. The figure A was prepared using PyMol (www.pymol.org), and figure B was generated by DS 3.0.

thromboembolic diseases.

5. Experimental section

5.1. Docking simulation

The protocols for docking simulation employing Glide in Schrodinger 2009 have been described in our published study [39]. fXa structure was obtained from the protein data bank (PDB code: 2w26) and prepared with water removed and hydrogen added at pH = 7.4. The active site was defined by the volume occupied by the native ligand. Protein structure was prepared using the "protein preparation wizard" to generate the grid file in the binding site. Other parameters were in default. The compounds for docking studies were drawn using ChemBioDraw Ultra 14.0 software and energy minimized using Prepare Ligands tool of DS3.0 and refined with CHARMm force field.

5.2. Chemistry

5.2.1. General chemistry

All starting materials, solvents and reagents were purchased from commercial sources and used without further purification. All the reactions mentioned in this study were monitored by thin-layer chromatography (TLC) at 254 nm under a UV lamp with the following eluent system: dichloromethane/methanol or *n*-hexane/ethyl acetate. Column chromatography separations were performed on silica gel (200–300 mesh) eluting with *n*-hexane/ethyl acetate (5/1 to 3/1) or dichloromethane/methanol (100/1 or 60/1). %Purity of the target compounds (>95%) were confirmed by HPLC analysis (UV detector, wavelength: 254 nm. Column: GL Sciences Inc. Inertsil ODS-SP C18 (4.6 mm × 250 mm, 5 μm), mobile phase: 100%MeCN, 0.1% TFA, flow rate: 1 mL/min). ^{1}H NMR and ^{13}C NMR spectra were recorded on a Bruker AV 300 MHz spectrometer using DMSO-d_6 or $CDCl_3$. Chemical shifts are expressed in δ (ppm) units relative to TMS and coupling constants (J) are recorded in Hertz (Hz). High resolution mass spectrometry (HRMS): Waters UPLC/MS, in *m/z*; Elemental analyses were carried out on CHN-O-Rapid instrument.

5.2.2. 1-(4-aminophenyl)piperidin-2-one (**2**, X = CH_2)

Compound **1** (5 g, 22.83 mmol) was dissolved in DMSO (120 mL), followed by the addition of 2-Piperidinone (4.53 g, 45.66 mmol), CuI (0.43 g, 2.28 mmol), 8-hydroxy-quinoline (0.66 g, 4.57 mmol) and K_2CO_3 (9.46 g, 68.49 mmol). The mixture was heated to 130 °C under N_2 for 12 h, cooled, and quenched with water (120 mL). The organics were extracted with ethyl acetate (2 × 150 mL) and dried (Na_2SO_4). Purification by silica gel column chromatography (*n*-hexane/ethyl acetate, 5/1 to 3/1, as eluent) afforded compound **2** as faint yellow (2.80 g, 64.47%). m.p. 118–120 °C; ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.00 (d, J = 8.76 Hz, 2 H), 6.56 (d, J = 8.76 Hz, 2 H), 3.56 (t, J = 5.44 Hz, 2 H), 2.52 (t, J = 5.24 Hz, 2 H), 1.92–1.89 (m, 4 H).

5.2.3. 4-(4-aminophenyl)morpholin-3-one (**2**, X = O)

The intermediate **2** (X = O) was prepared according to the procedure outlined for compound 2 (X = CH_2). Colorless solid (3.1 g, 53.45%). m.p. 115–128 °C; ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.07 (d, J = 8.8 Hz, 2 H), 6.69 (d, J = 8.8 Hz, 2 H), 4.32 (s, 2 H), 4.00 (t, J = 4.8 Hz, 2 H), 3.72 (brs, 2 H), 3.69 (t, J = 4.8 Hz, 2 H).

5.2.4. 5-Methoxy-2-nitro-N-(4-(2-oxopiperidin-1-yl)phenyl) benzamide (**3**, X = CH_2)

To a solution of 5-methoxy-2-nitrobenzoic acid (18.5 g, 94.6 mmol) in 90 mL dichloromethane at 0 °C was added five drops of DMF. Oxalyl chloride (15.0 g, 119.1 mmol) was added dropwise. The mixture was stirred for 1 h at room temperature then solvent was removed under reduced pressure. The resultant oil was dissolved in 100 mL dichloromethane and this solution was added to a solution of **2** (15 g, 78.9 mmol) and triethylamine (21.98 mL, 15.96 g, 157.7 mmol) in 105 mL dichloromethane at 0 °C. The mixture was allowed to warm to room temperature overnight (16 h). The precipitate was then filtered and washed with water (20 mL) and dichloromethane (10 mL), and dried in a vacuum to afford a colorless solid (25.89, 88.9%), m.p. 165–167 °C;^{1}H NMR (300 MHz, DMSO-d_6): δ 10.74 (s, 1H), 8.18 (s, 1H), 7.65–7.68 (d, J = 8.76 Hz, 2H), 7.34–7.37 (d, J = 8.76 Hz, 2H), 7.21–7.24 (m, J = 8.6 Hz, 2H), 3.88 (s, 3H), 3.58 (t, J = 5.7 Hz, 2H), 2.38 (t, J = 6.0 Hz, 2H), 1.85 (m, 4H).

5.2.5. 5-Methoxy-2-nitro-N-(4-(3-oxomorpholino)phenyl) benzamide (**3**, X = O)

Compound **3** (X = O) was prepared from 5-methoxy-2-nitrobenzoic acid and 4-(4-amino- phenyl)morpholin-3-one **2** in the same method as that described for **3** (X = CH_2). Colorless solid (7.35 g, 76.09%). m.p.171–175 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 10.79 (s, 1H), 8.18–8.21 (m, 1H),7.68 (d, J = 8.76 Hz, 2H),7.38 (d, J = 8.76 Hz, 2H), 7.22–7.25 (m, J = 8.6 Hz, 2H), 4.20 (s, 2H), 3.94 (s, 3H), 3.94–3.99 (m, 2H), 3.71–3.74 (m, 2 H).

5.2.6. 2-Amino-5-methoxy-N-(4-(2-oxopiperidin-1-yl)phenyl) benzamide (**4**, X = CH_2)

To a solution of **3** (X = CH_2, 10.9 g, 90.4 mmol) in ethanol (100 mL) was added Pd/C (10%, 1.0 g). Then hydrazine hydrate (80%, 7.39 g) was added. After addition was complete, the mixture was gently heated at the reflux temperature for 2.0 h. The reaction mixture was cooled, and dichloromethane (50 ml) was added. The catalyst was filtered off through celite, and the filtrate was concentrated to afford the desired product 8 as a greenyellow solid (8.4 g, 83.87%). m.p.141–144 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 10.06 (s, 1H). 7.73 (d, J = 8.79 Hz, 2H), 7.33 (d, J = 8.79 Hz, 2H, Ar-H), 7.16 (s, 1H,), 6.91 (m, 1H), 6.70–6.73 (m, 1H), 5.88 (brs, 2H), 3.83 (s, 3H), 3.56 (t, J = 5.7 Hz, 2H), 2.37 (t, J = 6.0 Hz, 2H), 1.84 (m, 4H).

5.2.7. 2-Amino-5-methoxy-N-(4-(3-oxomorpholino)phenyl) benzamide(**4**, X = O)

Compound **4** (X = O) was prepared from 5-methoxy-2-nitro-*N*-(4-(3-oxomorpholino) phenyl) benzamide **3** (X = O, 6.0 g, 16.16 mmol) in the same method as that described for **4** (X = CH_2) as a colorless solid (5.23, 94.82%). m.p. 148–150 °C; ^{1}H NMR (300 MHz, DMSO-$d_{6\ 6}$): δ 10.08 (s, 1H). 7.70–7.73 (d, J = 8.79, 2H), 7.33–7.36 (d, J = 8.79, 2H, Ar-H), 7.18 (s, 1H,), 6.90–6.94 (m, 1H), 6.71–6.74 (m, 1H), 6.71–6.74 (m, 1H), 5.88 (brs, 2H), 4.20 (s, 2H), 3.94–3.99 (m, 2H,), 3.86 (s, 3H), 3.70–3.73 (m, 2H).

5.2.8. 6-Methoxy-3-(4-(2-oxopiperidin-1-yl)phenyl)-2,3-dihydroquinazolin-4(1H)-one (**5**, X = CH_2)

To a solution of 4 (X = CH_2, 5 g, 14.73 mmol) in 30 mL ethanol at room temperature was added NaOH (1.30 g, 32.41). The solution was heated to 37 °C, and treated with 37% HCHO (1.21 mL, 16.21 mmol) over 5 min. The mixture was stirred at this temperature for 15 min. The resulting precipitate was filtered, washed with 8 mL H_2O/EtOH (v/v, 1:1), and dried (Na_2SO_4), affording the desired product as a bright yellow solid (4.01 g, 77.46%). m.p. 149–152 °C; ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.58 (s, 1H). 7.38–7.41 (d, J = 8.79 Hz, 2H). 7.32–7.35 (d, J = 8.79 Hz, 2H). 7.03 (s, 1H). 6.83 (s, 1H). 4.93 (s, 2H). 3.85 (s, 3H), 3.67 (t, J = 5.7 Hz, 2H), 2.59 (t, J = 5.7 Hz, 2H). 1.98 (m, 4H).

*5.2.9. 4-(4-(6-methoxy-4-oxo-1,2-dihydroquinazolin-3(4H)-yl) phenyl)morpholin-3-one (**5**, X = O)*

Compound **5** (X = O) was prepared from **4** (X = O, 6.0 g, 17.58 mmol) in the same method as that described for **5** (X = CH_2) as a colorless solid (3.98 g, 64.08%). m.p. 156–159 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.49–7.35 (m, 4H), 7.28 (d, *J* = 2.8 Hz, 1H), 7.02 (dd, *J* = 8.7, 3.0 Hz, 1H), 6.83 (d, *J* = 8.8 Hz, 1H), 6.59 (s, 1H), 4.90 (d, *J* = 2.7 Hz, 2H), 4.21 (s, 2H), 4.04–3.95 (m, 2H), 3.80–3.66 (m, 5H).

*5.2.10. 1-(4-aminophenyl)pyridin-2(1H)-one (**9**)*

The intermediate **9** was prepared according to the procedure outlined for compound **2**. Colorless solid (3.1 g, 53.45%). m.p. 128–130 °C ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.53 (dd, *J* = 6.8, 1.5 Hz, 1H), 7.44 (ddd, *J* = 8.8, 6.6, 2.1 Hz, 1H), 6.98 (d, *J* = 8.6 Hz, 2H), 6.61 (d, *J* = 8.6 Hz, 2H), 6.41 (d, *J* = 9.1 Hz, 1H), 6.23 (td, *J* = 6.7, 1.1 Hz, 1H), 5.32 (s, 2H).

*5.2.11. 5-Methoxy-2-nitro-N-(4-(2-oxopyridin-1(2H)-yl)phenyl) benzamide(**10**)*

The intermediate **10** was prepared from **9** following the protocols employed for compound **3**. Faint yellow solid (1.5 g, 43.21%). m.p. 171–173 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 10.76 (s, 1H), 8.22 (d, *J* = 8.8 Hz, 1H), 7.77 (d, *J* = 8.8 Hz, 2H), 7.64 (dd, *J* = 6.8, 1.4 Hz, 1H), 7.55–7.45 (m, 1H), 7.39 (d, *J* = 8.7 Hz, 2H), 7.33–7.19 (m, 2H), 6.48 (d, *J* = 9.1 Hz, 1H), 6.32 (t, *J* = 6.6 Hz, 1H), 3.95 (s, 3H).

*5.2.12. 2-Amino-5-methoxy-N-(4-(2-oxopyridin-1(2H)-yl)phenyl) benzamide(**11**)*

The title compound was synthesized from **10** in the same procedure as that described for **4**. Colorless solid (1.3 g, 71%). m.p. 149–151 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 10.17 (s, 1H), 7.81 (d, *J* = 8.4 Hz, 2H), 7.64 (d, *J* = 5.7 Hz, 1H), 7.55–7.45 (m, 1H), 7.36 (d, *J* = 8.3 Hz, 2H), 7.20 (s, 1H), 6.94 (d, *J* = 7.2 Hz, 1H), 6.74 (d, *J* = 8.6 Hz, 1H), 6.47 (d, *J* = 9.4 Hz, 1H), 6.31 (t, *J* = 6.5 Hz, 1H), 5.90 (s, 2H), 3.74 (s, 3H).

*5.2.13. 6-Methoxy-3-(4-(2-oxopyridin-1(2H)-yl)phenyl)-2,3-dihydroquinazolin-4(1H)-one(**12**)*

The title intermediate was prepared from **11** according to the procedure for preparation of **5**. Bright green solid (0.65 g, 69.12%). m.p. 163–165 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.67 (d, *J* = 5.8 Hz, 1H), 7.58–7.38 (m, 5H), 7.30 (d, *J* = 2.6 Hz, 1H), 7.04 (dd, *J* = 8.7, 2.8 Hz, 1H), 6.85 (d, *J* = 8.8 Hz, 1H), 6.63 (s, 1H), 6.49 (d, *J* = 9.3 Hz, 1H), 6.33 (t, *J* = 6.6 Hz, 1H), 4.95 (d, *J* = 2.3 Hz, 2H), 3.73 (s, 3H).

*5.2.14. 1-(4-chlorobenzoyl)-6-methoxy-3-(4-(2-oxopiperidin-1-yl) phenyl)-2,3-dihydroquinazolin-4(1H)-one(**X138**)*

To a solution of 4-chlorobenzoic acid (0.11 g, 0.68 mmol) in 2 mL dichloromethane at 0 °C was added two drops of DMF. Oxalyl chloride (0.26 g, 0.17 mL, 2.05 mmol) was added dropwise. The mixture was stirred for 1 h at room temperature then solvent was removed in vacuo. The resultant oil was dissolved in 3 mL dichloromethane and this solution was added to a solution of 11 (0.2 g, 0.57 mmol) and triethylamine (86.39 mg, 0.12 mL, 0.85 mmol) in 3 mL dichloromethane at 0 °C. The mixture was allowed to warm to room temperature overnight (14 h). The mixture was washed with purified water and saturated salt water. The organic layer was dried over sodium sulfate and concentrated in vacuo. The resulted residue was chromatographed on silica gel eluting with CH_2Cl_2/MeOH (60/1–30/1) to give the desired compound **X138** as white solid (176 mg, 63.1%.). m.p. 239–243 °C; ^{1}H NMR (300 MHz, DMSO-d_6,) δ: 7.48 (d, *J* = 8.8 Hz, 5H), 7.36 (dd, *J* = 18.0, 8.7 Hz, 4H), 7.07 (s, 2H), 5.54 (s, 2H), 3.82 (s, 3H), 3.62 (s, 2H), 2.41 (d, *J* = 5.9 Hz, 2H), 1.86 (s, 4H); ^{13}C NMR (75 MHz, DMSO-d_6): δ 168.95, 156.86, 141.88, 137.77, 135.76, 133.21, 132.76, 130.32, 128.65, 126.72, 125.50, 125.14, 119.25, 111.29, 62.72, 55.63, 50.75, 32.61, 23.00, 20.91; Anal. calcd. for $C_{27}H_{24}ClN_3O_4$ (100%): C, 66.19; H, 4.94; N, 8.58. Found (100%): C, 66.26; H, 4.85; N, 8.65; HR-MS (ESI) *m*/*z*: calculated for $C_{27}H_{24}ClN_3O_4$ $[M+H]^+$: 490.1455, found: 490.1454.

*5.2.15. 1-Benzoyl-6-methoxy-3-(4-(2-oxopiperidin-1-yl)phenyl)-2,3-dihydroquinazolin-4(1H)-one(**6a**)*

Compound **6a** was prepared from **5** (X = CH_2, 0.20 g, 0.57 mmol) and benzoic acid (0.08 g, 0.68 mmol) in the same method as that described for **X138** as white solid (156 mg, 55.9%). M.p. 223–228 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.62–7.41 (m, 6H), 7.40–7.26 (m, 4H), 7.06 (s, 2H), 5.53 (s, 2H), 3.82 (s, 3H), 3.61 (s, 2H), 2.40 (s, 2H), 1.86 (s, 4H); ^{13}C NMR (75 MHz, DMSO): δ 168.93, 161.37, 156.77, 141.86, 137.75, 133.95, 133.43, 131.10, 128.57, 128.14, 126.71, 125.43, 125.18, 124.24, 119.27, 111.15, 62.62, 55.62, 50.74, 32.61, 22.99, 20.90; Anal. calcd. for $C_{27}H_{25}N_3O_4$ (%): C, 71.19; H, 5.53; N, 9.22. Found (%): C, 71.06; H, 5.67; N, 9.10; HR-MS (ESI) *m*/*z*: calculated for $C_{27}H_{25}N_3O_4$ $[M+H]^+$: 456.1845, found: 456.1835.

*5.2.16. 6-Methoxy-1-(4-methoxybenzoyl)-3-(4-(2-oxopiperidin-1-yl)phenyl)-2,3-dihydroquinazolin-4(1H)-one(**6b**)*

The title compound was prepared according to the procedure outlined for compound **X138** as white solid (125 mg, 49.3%). m.p. 233–236 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ7.58–7.41 (m, 3H), 7.34 (q, *J* = 8.8 Hz, 4H), 7.14–7.00 (m, 2H), 6.95 (d, *J* = 8.8 Hz, 2H), 5.55 (s, 2H), 3.82 (s, 3H), 3.79 (s, 3H), 3.62 (s, 2H), 2.40 (d, *J* = 6.0 Hz, 2H), 1.85 (s, 4H); ^{13}C NMR (75 MHz, DMSO): δ 168.89, 168.05, 161.40, 156.56, 141.78, 137.76, 133.88, 130.53, 126.65, 125.65, 125.32, 125.03, 123.93, 119.34, 113.78, 111.02, 62.73, 55.55, 55.33, 50.69, 32.56, 22.95, 20.85; Anal. calcd. for $C_{28}H_{27}N_3O_5$ (%): C, 69.26; H, 5.61; N, 8.65. Found (%): C, 70.06; H, 5.75; N, 8.55; HR-MS (ESI) *m*/*z*: calculated for $C_{28}H_{27}N_3O_5$ $[M+H]^+$: 486.1951, found: 486.1961.

*5.2.17. 1-(5-Chlorothiophene-2-carbonyl)-6-methoxy-3-(4-(2-oxopiperidin-1-yl)phenyl)-2,3-dihydroquinazolin-4(1H)-one(**6c**)*

The title compound was prepared according to the procedure outlined for compound **X138** as colorless solid (131 mg, 64.7%). m.p. 230–232 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.48 (d, *J* = 2.9 Hz, 1H), 7.43–7.20 (m, 5H), 7.20–7.05 (m, 3H), 5.64 (s, 2H), 3.85 (s, 3H), 3.62 (s, 2H), 2.41 (s, 2H), 1.86 (s, 4H).; ^{13}C NMR (75 MHz, DMSO-d_6): δ 168.93, 161.21, 160.55, 157.38, 141.87, 137.65, 134.82, 134.66, 132.81, 132.56, 127.64, 126.73, 125.51, 125.44, 124.66, 119.41, 111.21, 62.83, 55.68, 50.75, 32.60, 23.00, 20.91; MS (ESI, *m*/*z*): 496.1 $(M+H)^+$, 518.1 $(M+Na)^+$; Anal. calcd. for $C_{25}H_{22}ClN_3O_4S$ (%): C, 60.54; H, 4.47; N, 8.47. Found (%): C, 60.49; H, 4.51; N, 8.52; HR-MS (ESI) *m*/*z*: calculated for $C_{25}H_{22}ClN_3O_4S$ $[M+H]^+$: 496.1020, found: 496.1021.

*5.2.18. 6-Methoxy-1-(3-methoxybenzoyl)-3-(4-(2-oxopiperidin-1-yl)phenyl)-2,3-dihydroquinazolin-4(1H)-one(**6d**)*

The title compound was prepared according to the procedure outlined for compound **X138** as white solid (0.12 mg, 71.2%). m.p. 223–228 °C; ^{1}H NMR (300 MHz, $CDCl_3$) δ: 7.46 (d, *J* = 2.9 Hz, 1H), 7.34 (q, *J* = 9.0 Hz, 5H), 7.05 (dd, *J* = 25.1, 10.5 Hz, 5H), 5.51 (s, 2H), 3.82 (s, 3H), 3.71 (s, 3H), 3.60 (d, *J* = 5.6 Hz, 2H), 3.30 (s, 2H), 2.41 (d, *J* = 6.1 Hz, 2H), 1.86 (d, *J* = 2.6 Hz, 4H); ^{13}C NMR (75 MHz, DMSO-d_6): δ 168.90, 168.20, 161.35, 159.06, 156.77, 141.82, 137.68, 135.31, 133.33, 129.76, 126.64, 125.37, 125.07, 124.23, 120.00, 119.25, 116.85, 113.13, 111.11, 62.71, 55.59, 55.26, 50.71, 32.58, 22.96, 20.86; Anal. calcd. for $C_{28}H_{27}N_3O_5$ (%): C, 69.26; H, 5.61; N, 8.65. Found (%): C, 69.18; H, 5.66; N, 8.71; HR-MS (ESI) *m*/*z*: calculated for $C_{28}H_{27}N_3O_5$ $[M+H]^+$: 486.1951, found: 486.1958.

*5.2.19. 1-(3-chlorobenzoyl)-6-methoxy-3-(4-(2-oxopiperidin-1-yl) phenyl)-2,3-dihydroquinazolin-4(1H)-one(**6e**)*

The title compound was prepared according to the procedure outlined for compound **X138** as white solid (0.10 mg, 41.9%). m.p. 230–235 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ7.60–7.55 (m, 2H), 7.47–7.31 (m, 7H), 7.09 (s, 2H), 5.51 (s, 2H), 3.82 (s, 3H), 3.61 (d, J = 5.7 Hz, 2H), 2.41 (d, J = 6.3 Hz, 2H), 1.86 (s, 4H); ^{13}C NMR (75 MHz, DMSO-d_6): δ 168.95, 161.32, 156.92, 141.89, 137.75, 136.08, 133.25, 133.06, 130.89, 130.47, 128.00, 126.70, 125.53, 125.12, 124.38, 119.21, 111.32, 62.66, 55.64, 50.76, 32.61, 23.00, 20.91; Anal. calcd. for $C_{27}H_{24}ClN_3O_4$ (%): C, 66.19; H, 4.94; N, 8.58. Found (%): C, 66.11; H, 4.98; N, 8.66; HR-MS (ESI) m/z: calculated for $C_{27}H_{24}ClN_3O_4$ $[M+H]^+$: 490.1455, found: 490.1456.

*5.2.20. 1-(2-(4-chlorophenyl)acetyl)-6-methoxy-3-(4-(2-oxopiperidin-1-yl)phenyl)-2,3-dihydroquinazolin-4(1H)-one(**X254**)*

To a solution of 2-(4-chlorophenyl) acetic acid (0.08 g, 0.47 mmol) and **5** (X = CH_2, 0.15 g, 0.43 mmol) in 2 mL DMF at 0 °C was added EDCI (0.11 g, 0.55 mmol) in 0.5 mL DMF. Then DMAP (5.1 mg, 0.04 mmol) was added to the reaction. The mixture was stirred for 24 h at room temperature. The reaction was quenched by addition of 3 mL cold water and extracted with DCM (3 × 3 mL). The combined organic layers were washed with 5 mL water 5 mL brine, dried over anhydrous Na_2SO_4, and filtered, and then the filtrate was concentrated in vacuo. The resulted residue was chromatographed on silica gel eluting with CH_2Cl_2/MeOH (60/1–30/1) to give the desired compound **X254** as white solid (112 mg, 55.78%.). m.p. 218–221 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.71 (d, J = 8.8 Hz, 1H), 7.41 (d, J = 3.0 Hz, 1H), 7.30 (m, 4H), 7.28 (m, 3H) 7.15 (d, J = 8.3 Hz, 2H), 5.52 (s, 2H), 4.01 (s, 2H), 3.84 (s, 3H), 3.62 (d, J = 5.8 Hz, 2H), 2.40 (t, J = 5.9 Hz, 2H), 1.87 (s, 4H). ^{13}C NMR (75 MHz, DMSO-d_6): δ 168.88, 159.56, 141.60, 137.62, 133.12, 130.98, 128.11, 126.51, 125.40, 125.21, 119.23, 55.61, 50.75, 32.54, 22.96, 20.87; Anal. calcd. for $C_{28}H_{26}ClN_3O_4$ (%): C, 66.73; H, 5.20; Cl, 7.03; N, 8.34. Found (%): C, 66.85; H, 5.28; N, 8.21; HR-MS (ESI) m/z: calculated for $C_{28}H_{26}ClN_3O_4$ $[M+H]^+$: 504.1612, found: 504.1614.

*5.2.21. 6-Methoxy-3-(4-(2-oxopiperidin-1-yl)phenyl)-1-(2-phenylacetyl)-2,3-dihydroquinazolin-4(1H)-one(**7a**)*

The title compound was prepared according to the procedure outlined for compound **X254** as white solid (0.12 g, 76.1%). m.p. 205–209 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.70 (d, J = 8.8 Hz, 1H), 7.39 (m, 2H), 7.26 (m, 5H), 7.14 (m, 2H), 6.80–6.71 (m, 1H), 6.66 (s, 1H), 5.51 (s, 2H), 3.96 (s, 2H), 3.83 (s, 3H), 3.61 (d, J = 5.8 Hz, 2H), 2.40 (t, J = 6.0 Hz, 2H), 1.97–1.75 (m, 4H). ^{13}C NMR (75 MHz, DMSO-d_6): δ 168.86, 159.87, 141.64, 135.86, 133.22, 129.30, 126.52, 125.30, 119.23, 114.42, 112.25, 55.60, 54.82, 50.72, 32.54, 22.95, 20.86; Anal. calcd. for $C_{28}H_{27}N_3O_4$ (%): C, 71.62; H, 5.80; N, 8.95. Found (%): C, 71.46; H, 5.67; N, 8.89; HR-MS (ESI) m/z: calculated for $C_{28}H_{27}N_3O_4$ $[M+H]^+$: 470.2002, found: 470.2004.

*5.2.22. 6-Methoxy-1-(2-(4-methoxyphenyl)acetyl)-3-(4-(2-oxopiperidin-1-yl)phenyl)-2,3-dihydroquinazolin-4(1H)-one(**7b**)*

The title compound was prepared according to the procedure outlined for compound **X254** as white solid (0.14 g, 51.4%). m.p. 225–228 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.69 (d, J = 8.8 Hz, 1H), 7.43–7.19 (m, 6H), 7.01 (d, J = 8.4 Hz, 2H), 6.77 (d, J = 8.4 Hz, 2H), 5.51 (s, 2H), 3.91 (s, 2H), 3.83 (s, 3H), 3.69 (s, 3H), 3.61 (d, J = 5.7 Hz, 2H), 2.39 (t, J = 6.0 Hz, 2H), 1.86 (s, 4H); ^{13}C NMR (75 MHz, DMSO-d_6): δ 168.39, 160.35, 159.91, 141.65, 135.62, 133.18, 131.05, 128.56, 126.56, 125.81, 112.03, 56.51, 55.81, 50.55, 32.56, 22.93, 20.87; Anal. calcd. for $C_{29}H_{29}N_3O_5$ (%): C, 69.72; H, 5.85; N, 8.41. Found (%): C, 69.78; H, 5.75; N, 8.46; HR-MS (ESI) m/z: calculated for $C_{29}H_{29}N_3O_5$ $[M+H]^+$: 500.2107, found: 500.2106.

*5.2.23. 1-(2-(2,4-dimethoxyphenyl)acetyl)-6-methoxy-3-(4-(2-oxopiperidin-1-yl)phenyl)-2,3-dihydroquinazolin-4(1H)-one(**7c**)*

The title compound was prepared according to the procedure outlined for compound **X254** as white solid (0.09 g, 39.8%). m.p. 238–241 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.59–7.22 (m, 6H), 7.15 (s, 1H), 6.61 (s, 3H), 5.47 (s, 2H), 3.82 (s, 3H), 3.66 (s, 6H), 3.61 (s, 2H), 2.39 (s, 2H), 1.85 (s, 4H). ^{13}C NMR (75 MHz, DMSO-d_6): δ 168.86, 160.30, 156.75, 141.78, 126.60, 125.32, 124.99,124.85, 124.25, 119.25, 111.03, 105.53, 102.67, 62.61, 55.57, 55.39, 50.69, 32.54, 22.93, 20.83; Anal. calcd. for $C_{30}H_{31}N_3O_6$ (%): C, 68.04; H, 5.90; N, 7.93. Found (%): C, 68.14; H, 5.85; N, 7.95; HR-MS (ESI) m/z: calculated for $C_{30}H_{31}N_3O_6$ $[M+H]^+$: 530.2213, found: 530.2223.

*5.2.24. 1-(2-(3-Chlorophenyl)acetyl)-6-methoxy-3-(4-(2-oxopiperidin-1-yl)phenyl)-2,3-dihydroquinazolin-4(1H)-one(**7d**)*

The title compound was prepared according to the procedure outlined for compound **X254** as white solid (0.15 g, 63.4%). m.p. 215–219 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.72 (d, J = 8.9 Hz, 1H), 7.41 (m, 1H), 7.32 (s, 4H), 7.26 (d, J = 5.0 Hz, 4H), 7.09 (s, 1H), 5.52 (s, 2H), 4.03 (s, 2H), 3.84 (s, 3H), 3.61 (d, J = 5.7 Hz, 2H), 2.40 (t, J = 5.9 Hz, 2H), 1.86 (s, 4H). ^{13}C NMR (75 MHz, DMSO-d_6): δ 168.87, 160.11, 141.66, 137.05, 133.10, 132.68, 129.94, 129.16, 127.99, 126.56, 125.37, 119.21, 55.62, 50.74, 32.55, 22.96, 20.86; Anal. calcd. for $C_{28}H_{26}ClN_3O_4$ (%): C, 66.73; H, 5.20; N, 8.34. Found (%): C, 66.79; H, 5.25; N, 8.29; HR-MS (ESI) m/z: calculated for $C_{28}H_{26}ClN_3O_4$ $[M+H]^+$: 504.1612, found: 504.1613.

*5.2.25. 1-(2-(2-Chlorophenyl)acetyl)-6-methoxy-3-(4-(2-oxopiperidin-1-yl)phenyl)-2,3-dihydroquinazolin-4(1H)-one(**7e**)*

The title compound was prepared according to the procedure outlined for compound **X254** as white solid (0.16 g, 65.7%). m.p. 226–229 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.69 (d, J = 8.9 Hz, 1H), 7.55 (s, 1H), 7.47 (d, J = 2.7 Hz, 1H), 7.40 (s, 3H), 7.36 (d, J = 5.2 Hz, 5H), 5.55 (s, 2H), 4.07 (s, 2H), 3.86 (s, 3H), 3.62 (s, 2H), 2.40 (t, J = 6.0 Hz, 2H), 1.86 (d, J = 2.3 Hz, 4H). ^{13}C NMR (75 MHz, DMSO-d_6): δ 169.01, 168.86, 162.41, 157.35, 141.66, 137.05, 133.10, 132.68, 129.94, 129.16, 127.99, 126.56, 125.37, 119.21, 55.21, 54.75, 50.74, 32.55, 22.96, 20.86; Anal. calcd. for $C_{28}H_{26}ClN_3O_4$ (%): C, 66.73; H, 5.20; N, 8.34. Found (%): C, 66.76; H, 5.25; N, 8.37. HR-MS (ESI) m/z: calculated for $C_{28}H_{26}ClN_3O_4$ $[M+H]^+$: 504.1612, found: 504.1617.

*5.2.26. N-(4-(2-(6-methoxy-4-oxo-3-(4-(2-oxopiperidin-1-yl) phenyl)-3,4-dihydroquinazolin-1(2H)-yl)-2-oxoethyl)phenyl) acetamide(**7f**)*

The title compound was prepared according to the procedure outlined for compound **X254** as white solid (78.8 mg, 35.4%). m.p. 231–233 °C; ^{1}H NMR (300 MHz, DMSO-d_6) δ 9.93 (s, 1H), 7.69 (d, J = 8.7 Hz, 1H), 7.39 (m, 4H), 7.26 (m, 4H), 7.00 (d, J = 8.1 Hz, 2H), 5.50 (s, 2H), 3.92 (s, 2H), 3.83 (s, 3H), 3.61 (d, J = 5.6 Hz, 2H), 2.41 (d, J = 5.7 Hz, 2H), 2.02 (s, 3H), 1.86 (s, 4H). ^{13}C NMR (75 MHz, DMSO-d_6): δ 170.11, 168.83, 162.56, 158.43, 140.10, 136.95, 133.82, 132.65, 129.87, 129.43, 127.12, 125.36, 123.91, 119.26, 55.21, 54.75, 50.74, 32.58, 23.45, 22.81, 21.06; Anal. calcd. for $C_{30}H_{30}N_4O_5$ (%): C, 68.43; H, 5.74; N, 10.64. Found (%): C, 68.47; H, 5.76; N, 10.59; HR-MS (ESI) m/z: calculated for $C_{28}H_{26}ClN_3O_4$ $[M+H]^+$: 527.2216, found: 527.2227.

*5.2.27. 4-(4-(1-(4-(Chloromethyl)benzoyl)-6-methoxy-4-oxo-1,2-dihydroquinazolin-3(4H)-yl)phenyl)morpholin-3-one(**8a**)*

The title compound was prepared according to the procedure outlined for compound **X254** as white solid (120.1 mg, 62.3%). m.p. 221–225 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.58–7.28 (m, 9H), 7.25–6.91 (m, 2H), 5.55 (s, 2H), 4.79 (s, 2H), 4.21 (s, 2H), 3.98 (t, J = 5.1 Hz, 2H), 3.82 (s, 3H), 3.75 (t, J = 5.1 Hz, 2H). ^{13}C NMR

(75 MHz, DMSO-d_6): δ 167.98, 166.07, 161.38, 156.80, 140.53, 139.79, 138.05, 133.75, 133.34, 128.87, 128.63, 125.92, 125.56, 125.15, 124.18, 119.32, 111.18, 67.74, 63.48, 55.62, 48.87, 45.31; Anal. calcd. for $C_{27}H_{24}ClN_3O_5$ (%): C, 64.10; H, 4.78; N, 8.31. Found (%): C, 64.14; H, 4.75; N, 8.33. HR-MS (ESI) *m/z*: calculated for $C_{27}H_{24}ClN_3O_5$ $[M+H]^+$: 505.1404, found: 505.1406.

*5.2.28. 4-(4-(1-(3-(4-Chlorophenyl)propanoyl)-6-methoxy-4-oxo-1,4-dihydroquinazolin-3(2H)-yl)phenyl)morpholin-3-one(**8b**)*

The title compound was prepared according to the procedure outlined for compound **X254** as white solid (81.5 mg, 51.8%). m.p. 208–211 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.53–7.46 (m, 3H), 7.45–7.39 (m, 3H), 7.26 (d, *J* = 8.7 Hz, 2H), 7.24–7.19 (m, 1H), 7.17 (d, *J* = 8.7 Hz, 2H), 5.51 (s, 2H), 4.23 (s, 2H), 4.0 (m, 2H), 3.83 (s, 3H), 3.79–3.75 (m, 2H), 2.91 (t, *J* = 7.6 Hz, 2H). 2.65 (t, *J* = 7.6 Hz, 2H); ^{13}C NMR (75 MHz, DMSO): δ 171.56, 166.13, 161.40, 157.21, 139.77, 139.70, 138.13, 133.25, 130.67, 130.23, 128.19, 125.92, 125.44, 119.30, 111.66, 67.79, 63.54, 55.65, 48.97, 34.23, 29.78; Anal. calcd. for $C_{28}H_{26}ClN_3O_5$ (%):C, 64.68; H, 5.04; N, 8.08. Found (%): C, 64.41; H, 5.15; N, 8.17; HR-MS (ESI) *m/z*: calculated for C28H26ClN3O5 $[M+H]^+$: 520.1561, found: 520.1562.

*5.2.29. 4-(4-(1-(2-(4-Chlorophenyl)acetyl)-6-methoxy-4-oxo-1,2-dihydroquinazolin-3(4H)-yl)phenyl)morpholin-3-one(**8c**)*

The title compound was prepared according to the procedure outlined for compound **X254** as white solid (61.7 mg, 38.6%). m.p. 235–238 °C; ^{1}H NMR (300 MHz, DMSO) δ 7.71 (d, *J* = 8.9 Hz, 1H), 7.51–7.35 (m, 4H), 7.27 (t, *J* = 6.5 Hz, 4H), 7.15 (d, *J* = 8.4 Hz, 2H), 5.53 (s, 2H), 4.22 (s, 2H), 3.99 (d, *J* = 8.9 Hz, 4H), 3.84 (s, 3H), 3.80–3.72 (m, 2H); ^{13}C NMR (75 MHz, DMSO-d_6): δ 166.05, 161.32, 157.20, 139.56, 137.98, 133.60, 133.19, 131.40, 131.03, 128.14, 125.77, 125.43, 125.32, 119.29, 111.47, 67.74, 63.49, 55.64, 48.93; Anal. calcd. for $C_{27}H_{24}ClN_3O_5$ (%): C, 64.10; H, 4.78; N, 8.31. Found (%): C, 64.06; H, 4.75; N, 8.36. HR-MS (ESI) *m/z*: calculated for $C_{27}H_{24}ClN_3O_5$ $[M+H]^+$: 505.1404, found: 505.1415.

*5.2.30. 4-(4-(6-Methoxy-1-(2-(4-methoxyphenyl)acetyl)-4-oxo-1,2-dihydroquinazolin-3(4H)-yl)phenyl)morpholin-3-one(**8d**)*

The title compound was prepared according to the procedure outlined for compound **X254** as white solid (101.1 mg, 62.5%). m.p. 230–233 °C; ^{1}H NMR (500 MHz, DMSO-d_6): δ 7.69 (d, *J* = 8.7 Hz, 1H), 7.46–7.23 (m, 6H), 7.01 (d, *J* = 7.8 Hz, 2H), 6.78 (d, *J* = 7.8 Hz, 2H), 5.52 (s, 2H), 4.22 (s, 2H), 4.01–3.97 (m, 2H), 3.91 (s, 2H), 3.84 (s, 3H), 3.79–3.75 (m, 2H), 3.70 (s, 3H). ^{13}C NMR (75 MHz, DMSO-d_6): δ 166.08, 161.34, 158.02, 157.10, 139.58, 138.01, 133.35, 130.39, 129.99, 126.28, 125.80, 125.45, 125.38, 119.33, 113.76, 113.67, 111.39, 67.76, 63.51, 55.65, 54.99, 48.94; Anal. calcd. for $C_{28}H_{27}N_3O_6$ (%): C, 67.06; H, 5.43; N, 8.38. Found (%): C, 67.11; H, 5.45; N, 8.32. HR-MS (ESI) *m/z*: calculated for $C_{28}H_{27}N_3O_6$ $[M+H]^+$: 502.1900, found: 505.1902.

*5.2.31. 4-(4-(1-(2-(5-Chlorothiophen-2-yl)acetyl)-6-methoxy-4-oxo-1,2-dihydroquinazolin-3(4H)-yl)phenyl)morpholin-3-one(**8e**)*

The title compound was prepared according to the procedure outlined for compound **X254** as white solid (82.1 mg, 46.4%). m.p. 215–219 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.70 (d, *J* = 8.9 Hz, 1H), 7.51–7.36 (m, 5H), 7.28 (dd, *J* = 8.8, 2.8 Hz, 1H), 6.89 (d, *J* = 3.6 Hz, 1H), 6.70 (d, *J* = 3.7 Hz, 1H), 5.55 (s, 2H), 4.23 (s, 4H), 4.04–3.96 (m, 2H), 3.86 (s, 3H), 3.81–3.73 (m, 2H). ^{13}C NMR (75 MHz, DMSO-d_6): δ 166.10, 161.36, 157.20, 139.69, 138.02, 135.30, 132.98, 127.16, 126.63, 125.89, 125.52, 125.35, 119.36, 111.62, 67.76, 63.51, 55.70, 48.96, 34.78; Anal. calcd. for $C_{25}H_{22}ClN_3O_5S$ (%): C, 58.65; H, 4.33; N, 8.21. Found (%): C, 58.66; H, 4.35; N, 8.29. HR-MS (ESI) *m/z*: calculated for $C_{25}H_{22}ClN_3O_5S$ $[M+H]^+$: 512.0969, found: 512.0968.

*5.2.32. 1-(2-(5-Chloropyridin-2-yl)acetyl)-6-methoxy-3-(4-(2-oxopiperidin-1-yl)phenyl)-2,3-dihydroquinazolin-4(1H)-one (**8f**)*

The title compound was prepared according to the procedure outlined for compound **X254** as white solid (41.6 mg, 15.1%). m.p. >250 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 8.51 (s, 1H), 7.91 (d, *J* = 8.9 Hz, 1H), 7.75 (*J* = 8.9 Hz, 1H), 7.46–7.42 (m, 6H), 5.54 (s, 2H), 4.22 (s, 2H), 4.20 (s, 2H), 3.99 (t, *J* = 4.4 Hz, 2H) 3.84 (s, 3H), 3.75 (t, *J* = 5.0 Hz, 2H). ^{13}C NMR (75 MHz, DMSO-d_6): δ 168.11, 163.51, 159.10, 139.71, 138.32, 135.800, 133.13, 127.46, 126.82, 126.19, 125.69, 125.55, 119.56, 111.93, 66.11, 63.82, 56.54, 48.61, 34.85; Anal. calcd. for $C_{26}H_{23}ClN_4O_5$ (%): C, 61.60; H, 4.57; N, 11.05. Found (%): C, 61.86; H, 4.55; N, 11.13. HR-MS (ESI) *m/z*: calculated for $C_{26}H_{23}ClN_4O_5$ $[M+H]^+$: 507.1357, found: 507.1359.

*5.2.33. 4-(4-(1-(2-(3,4-Dichlorophenyl)acetyl)-6-methoxy-4-oxo-1,2-dihydroquinazolin-3(4H)-yl)phenyl)morpholin-3-one(**8g**)*

The title compound was prepared according to the procedure outlined for compound **X254** as white solid (100.1 mg, 53.6%). m.p. 228–232 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.72 (d, *J* = 9.3 Hz, 1H), 7.64–7.20 (m, 8H), 7.15 (d, *J* = 8.1 Hz, 1H), 5.53 (s, 2H), 4.22 (s, 2H), 4.15–3.92 (m, 4H), 3.84 (s, 3H), 3.77 (s, 2H). ^{13}C NMR (75 MHz, DMSO-d_6): δ 166.05, 161.32, 157.12, 139.59, 137.97, 135.87, 133.10, 131.44, 130.61, 130.20, 129.86, 129.34, 125.75, 125.46, 125.37, 119.26, 111.53, 67.74, 63.49, 55.67, 48.91; Anal. calcd. for $C_{27}H_{23}Cl_2N_3O_5$ (%): C, 60.01; H, 4.29; N, 7.78. Found (%): C, 60.06; H, 4.25; N, 7.75; HR-MS (ESI) *m/z*: calculated for $C_{27}H_{23}Cl_2N_3O_5$ $[M+H]^+$: 541.1015, found: 541.1025.

*5.2.34. 4-(4-(1-(2-(2,4-Dichlorophenyl)acetyl)-6-methoxy-4-oxo-1,2-dihydroquinazolin-3(4H)-yl)phenyl)morpholin-3-one(**8h**)*

The title compound was prepared according to the procedure outlined for compound **X254** as white solid (80.5 mg, 34.4%). m.p. 232–236 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.69 (d, *J* = 8.8 Hz, 1H), 7.55 (s, 1H), 7.52–7.35 (m, 7H), 7.29 (d, *J* = 8.8 Hz, 1H), 5.57 (s, 2H), 4.22 (s, 2H), 4.08 (s, 2H), 3.99 (t, *J* = 5.1 Hz, 2H), 3.86 (s, 3H), 3.77 (t, *J* = 5.1 Hz, 2H). ^{13}C NMR (75 MHz, DMSO-d_6): δ 166.07, 139.68, 138.03, 134.38, 133.49, 133.20, 132.44, 128.49, 127.27, 125.95, 125.48, 125.42, 119.45, 111.67, 67.74, 63.49, 55.70, 48.93, 37.46; Anal. calcd. for $C_{27}H_{23}Cl_2N_3O_5$: C, 60.01; H, 4.29; N, 7.78. Found: C, 60.10; H, 4.26; N, 7.74; HR-MS (ESI) *m/z*: calculated for $C_{27}H_{23}Cl_2N_3O_5$ $[M+H]^+$: 541.1015, found: 541.1018.

*5.2.35. 1-(2-(4-chlorophenyl)acetyl)-6-methoxy-3-(4-(2-oxopyridin-1(2H)-yl)phenyl)-2,3-dihydroquinazolin-4(1H)-one(**13**)*

The title compound was prepared according to the procedure outlined for compound **X138** as white solid (110.6 mg, 56.4%). m.p. >250 °C; ^{1}H NMR (300 MHz, DMSO-d_6): δ 7.77–7.63 (m, 2H), 7.58–7.34 (m, 6H), 7.27 (s, 3H), 7.15 (d, *J* = 8.1 Hz, 2H), 6.50 (d, *J* = 9.3 Hz, 1H), 6.34 (t, *J* = 6.5 Hz, 1H), 5.58 (s, 2H), 4.02 (s, 2H), 3.85 (s, 3H). ^{13}C NMR (75 MHz, DMSO-d_6): δ 161.31 (d, *J* = 8.3 Hz), 140.70, 139.62, 139.00, 138.53, 133.60, 133.25, 131.36, 131.07, 128.20, 127.17, 125.51, 125.37, 120.56, 119.43, 105.71, 55.70; Anal. calcd. for $C_{28}H_{22}ClN_3O_4$: C, 67.27; H, 4.44; N, 8.41. Found: C, 67.46; H, 4.41; N, 8.44; HR-MS (ESI) *m/z*: calculated for $C_{28}H_{22}ClN_3O_4$ $[M+H]^+$: 500.1299, found: 500.1295.

5.3. Inhibition of fXa and thrombin in vitro

The synthesized compounds were measured *in vitro* for their factor Xa/thrombin inhibition using achromogenic substrates. The protocols have been reported in our precious study [26]. The targeted compounds were prepared at 100 mM concentration and then serially diluted to give twelve different aliquots spanning a range of 0.010–500 μMin buffer (0.02 M Bis-Tris; 0.7 M NaCl; pH 6.0). The final enzyme concentrations were 0.003 U/mL (human fXa

from Enzyme Research Laboratories) and 0.125 U/mL (human thrombin from Sigma Chemical Co.). The reagents containing compound dilutions, buffer and enzyme were mixed, centrifuged, and incubated for 30 min at 37 °C in 96-well microtiter plates. The enzyme reaction was initiated by adding appropriate substrate (for factor Xa, S-2765; for thrombin, S-2238). The time course of the reaction was monitored continuously for 20 min at 405 nm in amicrotiter plate reader. The IC_{50} was calculated from the mean of duplicates from a dilution series of the compound with graphpad prism 5.

5.4. In vitro anticoagulant activity

2 × PT and 2 × aPTT were measured using commercially available kits. All tested Compounds or DMSO (3 μL) were added to 100 μL citrated rabbit plasma containing 3.8% sodium citrate, [1:9 (v/v)] and incubated for 10 min at 37 °C. Clotting times were measured in a coagulometer (A52 Semi-auto Coagulometer 2-channel analyzer) in accordance with the manufacturer's instructions and each test was done three times. Anticoagulant efficacy was defined as the concentration of inhibitor required to double PT and aPTT.

5.5. In vivo antithrombotic efficacy assays

An arteriovenous (AV) shunt in anesthetized rats (n = 10 per dose group) was performed as described previously with minor modifications [40,41]. The left jugular vein and right common carotid artery were cannulated with 8 cm long, saline-filled polyethylene tubing. Then the tube was connected by inserting a 5 cm long polyethylene tube containing rough nylon thread (50 × 0.15 mm) which was folded into a double string. **8e** or betrixaban dissolved in 0.5% carboxymethylcellulose sodium (CMC-Na) solution or vehicle was given orally 60 min before the shunt was opened for 15 min. The nylon thread was withdrawn rapidly, cleaned in saline, dried and weighed. The protocol for this experiment was approved by the Animal Care and Use Committee of China Pharmaceutical University (Nanjing, China).

5.6. Tail-bleeding model

Bleeding risk was evaluated in anesthetized rats according the reported protocols [42]. Compound 8e, betrixaban or vehicle was given orally 60 min before the tails of anesthetized rats were transected 2 mm from the tip and immersed vertically immediately into saline at 37 °C. The time until continuous blood flow ceased for >30 s was recorded with a maximum observation time of 30 min. The use of rats in this study was approved by the Animal Care and Use Committee of China Pharmaceutical University (Nanjing, China).

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.ejmech.2016.09.055.

References

[1] I.S.C.f.W.T. Day, Thrombosis: a major contributor to global disease burden, Thromb. Res. 134 (2014) 931–938.
[2] M. Germain, D.I. Chasman, H. de Haan, W. Tang, S. Lindstrom, L.C. Weng, M. de Andrade, M.C. de Visser, K.L. Wiggins, P. Suchon, N. Saut, D.M. Smadja, G. Le Gal, A. van Hylckama Vlieg, A. Di Narzo, K. Hao, C.P. Nelson, A. Rocanin-Arjo, L. Folkersen, R. Monajemi, L.M. Rose, J.A. Brody, E. Slagboom, D. Aissi, F. Gagnon, J.F. Deleuze, P. Deloukas, C. Tzourio, J.F. Dartigues, C. Berr, K.D. Taylor, M. Civelek, P. Eriksson, C. Cardiogenics, B.M. Psaty, J. Houwing-Duitermaat, A.H. Goodall, F. Cambien, P. Kraft, P. Amouyel, N.J. Samani, S. Basu, P.M. Ridker, F.R. Rosendaal, C. Kabrhel, A.R. Folsom, J. Heit, P.H. Reitsma, D.A. Tregouet, N.L. Smith, P.E. Morange, Meta-analysis of 65,734 individuals identifies TSPAN15 and SLC44A2 as two susceptibility loci for venous thromboembolism, Am. J. Hum. Genet. 96 (2015) 532–542.
[3] C.M. Yu, A.A. Khattab, S.C. Bertog, A.P. Lee, J.S. Kwong, H. Sievert, B. Meier, Mechanical antithrombotic intervention by LAA occlusion in atrial fibrillation, Nat. Rev. Cardiol. 10 (2013) 707–722.
[4] M. Levi, E.S. Eerenberg, P.W. Kampuisen, Anticoagulants. Old and new, Hamostaseologie 31 (2011) 229–235.
[5] S. Benmira, Z.K. Banda, V. Bhattacharya, Old versus new anticoagulants: focus on pharmacology, Recent Pat. Cardiovasc. Drug Discov. 5 (2010) 120–137.
[6] S. Srivastava, L.N. Goswami, D.K. Dikshit, Progress in the design of low molecular weight thrombin inhibitors, Med. Res. Rev. 25 (2005) 66–92.
[7] Y.K. Agrawal, H. Vaidya, H. Bhatt, K. Manna, P. Brahmkshatriya, Recent advances in the treatment of thromboembolic diseases: venous thromboembolism, Med. Res. Rev. 27 (2007) 891–914.
[8] Y.K. Lee, M.R. Player, Developments in factor Xa inhibitors for the treatment of thromboembolic disorders, Med. Res. Rev. 31 (2011) 202–283.
[9] D.L. Wolf, P.H. Lin, S. Hollenbach, A. Wong, D.R. Phillips, U. Sinha, Procoagulant activity of reversibly acylated human factor Xa, Blood 86 (1995) 4153–4157.
[10] D.J. Pinto, J.M. Smallheer, D.L. Cheney, R.M. Knabb, R.R. Wexler, Factor Xa inhibitors: next-generation antithrombotic agents, J. Med. Chem. 53 (2010) 6243–6274.
[11] J.I. Weitz, Factor Xa or thrombin: is thrombin a better target? Haemost. 1 (2007) 65–67.
[12] E. Perzborn, S. Roehrig, A. Straub, D. Kubitza, F. Misselwitz, The discovery and development of rivaroxaban, an oral, direct factor Xa inhibitor, Nat. Rev. Drug Dis. 10 (2011) 61–75.
[13] J.S. Paikin, J.J. Manolakos, J.W. Eikelboom, Rivaroxaban for stroke prevention in atrial fibrillation: a critical review of the ROCKET AF trial, Expert Rev. Cardiovasc. Ther. 10 (2012) 965–972.
[14] C. Frost, J. Wang, S. Nepal, A. Schuster, Y.C. Barrett, R. Mosqueda-Garcia, R.A. Reeves, F. LaCreta, Apixaban, an oral, direct factor Xa inhibitor: single dose safety, pharmacokinetics, pharmacodynamics and food effect in healthy subjects, Br. J. Clin. Pharmacol. 75 (2013) 476–487.
[15] P. Quon, H.H. Le, V. Raymond, M. Mtibaa, A. Moshyk, Clinical and economic benefits of extended treatment with apixaban for the treatment and prevention of recurrent venous thromboembolism in Canada, J. Med. Econ. (2016) 1–11.
[16] T. Furugohri, K. Isobe, Y. Honda, C. Kamisato-Matsumoto, N. Sugiyama, T. Nagahara, Y. Morishima, T. Shibano, DU-176b, a potent and orally active factor Xa inhibitor: in vitro and in vivo pharmacological profiles, J. Thromb. Haemost. 6 (2008) 1542–1549.
[17] M. Sherid, S. Sulaiman, S. Samo, H. Husein, R. Tupper, C. Spurr, H. Sifuentes, S. Sridhar, Risk of gastrointestinal bleeding with rivaroxaban: a comparative study with Warfarin, Gastroenterol. Res. Pract. 2016 (2016) 9589036.
[18] P. Konturek, T. Hess, Gastrointestinal bleeding under treatment with new oral anticoagulants, MMW Fortschr. Med. 156 (2014) 50–52.
[19] M.W. Sherwood, C.C. Nessel, A.S. Hellkamp, K.W. Mahaffey, J.P. Piccini, E.Y. Suh, R.C. Becker, D.E. Singer, J.L. Halperin, G.J. Hankey, S.D. Berkowitz, K.A. Fox, M.R. Patel, Gastrointestinal bleeding in patients with atrial fibrillation treated with rivaroxaban or warfarin: ROCKET AF trial, J. Am. Coll. Cardiol. 66 (2015) 2271–2281.
[20] Y.H. Mekaj, A.Y. Mekaj, S.B. Duci, E.I. Miftari, New oral anticoagulants: their advantages and disadvantages compared with vitamin K antagonists in the prevention and treatment of patients with thromboembolic events, Ther. Clin. Risk Manag. 11 (2015) 967–977.
[21] C.E. Dempfle, Direct oral anticoagulants–pharmacology, drug interactions, and side effects, Semin. Hematol. 51 (2014) 89–97.
[22] G. Lu, F.R. DeGuzman, S.J. Hollenbach, M.J. Karbarz, K. Abe, G. Lee, P. Luan, A. Hutchaleelaha, M. Inagaki, P.B. Conley, D.R. Phillips, U. Sinha, A specific antidote for reversal of anticoagulation by direct and indirect inhibitors of coagulation factor Xa, Nat. Med. 19 (2013) 446–451.
[23] E. Dolgin, Antidotes edge closer to reversing effects of new blood thinners, Nat. Med. 19 (2013) 251.
[24] D.M. Siegal, J.T. Curnutte, S.J. Connolly, G. Lu, P.B. Conley, B.L. Wiens, V.S. Mathur, J. Castillo, M.D. Bronson, J.M. Leeds, F.A. Mar, A. Gold, M.A. Crowther, Andexanet alfa for the reversal of factor Xa inhibitor activity, N. Engl. J. Med. 373 (2015) 2413–2424.
[25] R.A. Al-Horani, A.Y. Mehta, U.R. Desai, Potent direct inhibitors of factor Xa based on the tetrahydroisoquinoline scaffold, Eur. J. Med. Chem. 54 (2012) 771–783.
[26] J. Xing, L. Yang, H. Li, Q. Li, L. Zhao, X. Wang, Y. Zhang, M. Zhou, J. Zhou,

H. Zhang, Identification of anthranilamide derivatives as potential factor Xa inhibitors: drug design, synthesis and biological evaluation, Eur. J. Med. Chem. 95 (2015) 388–399.
[27] A.A. Khorana, Cancer and thrombosis: implications of published guidelines for clinical practice, Ann. Oncol. 20 (2009) 1619–1630.
[28] A.Y. Lee, Treatment of established thrombotic events in patients with cancer, Thromb. Res. 129 (Suppl. 1) (2012) S146–S153.
[29] A. Lee, VTE in patients with cancer–diagnosis, prevention, and treatment, Thromb. Res. 123 (Suppl. 1) (2008) S50–S54.
[30] J.S. Ventura, F. Faria, I.F. Batista, S.M. Simons, D.G. Oliveira, K.L. Morais, A.M. Chudzinski-Tavassi, A Kunitz-type FXa inhibitor affects tumor progression, hypercoagulable state and triggers apoptosis, Biomed. Pharmacother. 67 (2013) 192–196.
[31] P. Zhang, W. Huang, L. Wang, L. Bao, Z.J. Jia, S.M. Bauer, E.A. Goldman, G.D. Probst, Y. Song, T. Su, J. Fan, Y. Wu, W. Li, J. Woolfrey, U. Sinha, P.W. Wong, S.T. Edwards, A.E. Arfsten, L.A. Clizbe, J. Kanter, A. Pandey, G. Park, A. Hutchaleelaha, J.L. Lambing, S.J. Hollenbach, R.M. Scarborough, B.Y. Zhu, Discovery of betrixaban (PRT054021), N-(5-chloropyridin-2-yl)-2-(4- (N,N-dimethylcarbamimidoyl)benzamido)-5-methoxybenz amide, a highly potent, selective, and orally efficacious factor Xa inhibitor, Bioorg. Med. Chem. Lett. 19 (2009) 2179–2185.
[32] T. Xue, S. Ding, B. Guo, Y. Zhou, P. Sun, H. Wang, W. Chu, G. Gong, Y. Wang, X. Chen, Y. Yang, Design, synthesis, and structure-activity and structure-pharmacokinetic relationship studies of novel [6,6,5] tricyclic fused oxazolidinones leading to the discovery of a potent, selective, and orally bioavailable FXa inhibitor, J. Med. Chem. 57 (2014) 7770–7791.
[33] Y.M. Choi-Sledeski, R. Kearney, G. Poli, H. Pauls, C. Gardner, Y. Gong, M. Becker, R. Davis, A. Spada, G. Liang, V. Chu, K. Brown, D. Collussi, R. Leadley Jr., S. Rebello, P. Moxey, S. Morgan, R. Bentley, C. Kasiewski, S. Maignan, J.P. Guilloteau, V. Mikol, Discovery of an orally efficacious inhibitor of coagulation factor Xa which incorporates a neutral *P*1 ligand, J. Med. Chem. 46 (2003) 681–684.
[34] J.T. Kohrt, C.F. Bigge, J.W. Bryant, A. Casimiro-Garcia, L. Chi, W.L. Cody, T. Dahring, D.A. Dudley, K.J. Filipski, S. Haarer, R. Heemstra, N. Janiczek, L. Narasimhan, T. McClanahan, J.T. Peterson, V. Sahasrabudhe, R. Schaum, C.A. Van Huis, K.M. Welch, E. Zhang, R.J. Leadley, J.J. Edmunds, The discovery of (2R,4R)-N-(4-chlorophenyl)-N- (2-fluoro-4-(2-oxopyridin-1(2H)-yl) phenyl)-4-methoxypyrrolidine-1,2-dicarboxamide (PD 0348292), an orally efficacious factor Xa inhibitor, Chem. Biol. Drug Des. 70 (2007) 100–112.
[35] K.J. Filipski, J.T. Kohrt, A. Casimiro-Garcia, C.A. Van Huis, D.A. Dudley, W.L. Cody, C.F. Bigge, S. Desiraju, S. Sun, S.N. Maiti, M.R. Jaber, J.J. Edmunds, A versatile copper-catalyzed coupling reaction of pyridin-2(1H)-ones with aryl halides, Tetrahedron Lett. 47 (2006) 7677–7680.
[36] K. Ozaki, Y. Yamada, T. Oine, T. Ishizuka, Y. Iwasawa, Studies on 4(1H)-quinazolinones. 5. Synthesis and antiinflammatory activity of 4(1H)-quinazolinone derivatives, J. Med. Chem. 28 (1985) 568–576.
[37] C.A. Van Huis, A. Casimiro-Garcia, C.F. Bigge, W.L. Cody, D.A. Dudley, K.J. Filipski, R.J. Heemstra, J.T. Kohrt, R.J. Leadley Jr., L.S. Narasimhan, T. McClanahan, I. Mochalkin, M. Pamment, J.T. Peterson, V. Sahasrabudhe, R.P. Schaum, J.J. Edmunds, Exploration of 4,4-disubstituted pyrrolidine-1,2-dicarboxamides as potent, orally active factor Xa inhibitors with extended duration of action, Bioorg. Med. Chem. 17 (2009) 2501–2511.
[38] A.T. Cohen, R.A. Boyd, J.W. Mandema, L. Dicarlo, R. Pak, A. Investigators, An adaptive-design dose-ranging study of PD 0348292, an oral factor Xa inhibitor, for thromboprophylaxis after total knee replacement surgery, J. Thromb. Haemost. 11 (2013) 1503–1510.
[39] J. Xing, Q. Li, S. Zhang, H. Liu, L. Zhao, H. Cheng, Y. Zhang, J. Zhou, H. Zhang, Identification of dipeptidyl peptidase IV inhibitors: virtual screening, synthesis and biological evaluation, Chem. Biol. Drug Des. 84 (2014) 364–377.
[40] C.N. Berry, D. Girard, C. Girardot, S. Lochot, C. Lunven, C. Visconte, Antithrombotic activity of argatroban in experimental thrombosis in the rabbit, Semin. Thromb. Hemost. 22 (1996) 233–241.
[41] K. Sato, Y. Taniuchi, T. Kawasaki, F. Hirayama, H. Koshio, Y. Matsumoto, Relationship between the antithrombotic effect of YM-75466, a novel factor Xa inhibitor, and coagulation parameters in rats, Eur. J. Pharmacol. 347 (1998) 231–236.
[42] E. Perzborn, J. Strassburger, A. Wilmen, J. Pohlmann, S. Roehrig, K.H. Schlemmer, A. Straub, In vitro and in vivo studies of the novel antithrombotic agent BAY 59-7939–an oral, direct Factor Xa inhibitor, J. Thromb. Haemost. 3 (2005) 514–521.